THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

Dec 15, 2022 TIME: 2:30 p.m.-5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL NINE questions.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.
- Some formulae you may need:

For $n \geq 2$:

$$\int \cos^n \theta \, d\theta = \frac{1}{n} \cos^{n-1} \theta \sin \theta + \frac{n-1}{n} \int \cos^{n-2} \theta \, d\theta$$

$$\int \sin^n \theta \, d\theta = -\frac{1}{n} \sin^{n-1} \theta \cos \theta + \frac{n-1}{n} \int \sin^{n-2} \theta \, d\theta$$

1. (21 marks) In each multiple choice question below, some numbered choices are given and some are correct (wordings like is and are do not indicate the singularity /plurality of the number of correct choices). Answer the question by adding up the numbers of the correct choices. In case all choices are wrong, answer 0. Explanation is **NOT** required. Each question carries 3 marks.

EXAMPLE: Which of the following is correct?

(1) $1+2=3$.

(2) $1+1=2$.

(4) $1/0=1$.

(8) $1-1=0$. Answer: 11.

(a) Which of the following matrices is in reduced row echelon form?

(1) $\begin{bmatrix} 1 & 0 & 2 & 4 \\ 1 & 1 & 0 & 5 \\ 0 & 0 & 1 & 0 \end{bmatrix}$.

(2) $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

(4) The 2×3 zero matrix.

(8) The 4×4 identity matrix.

(b) For a homogeneous system of 5 linear equations with 4 unknowns, which of the following is possible?

(1) The solution set is empty.

(2) The system has a unique solution.

(4) The system has a unique non-trivial solution.

(8) The system has infinitely many solutions.

(c) Let A and B be square matrices of the same size. If A and B are both _____ , so is AB.

(1) diagonalizable.

(2) symmetric.

(4) upper triangular.

(8) non-singular.

(d) Let A be an $n \times n$ matrix. Which of the following is correct?

(1) Eigenvectors with distinct eigenvalues must be linearly independent.

(2) If A is a triangular matrix, then the diagonal elements of A are the eigenvalues of A.

(4) A and A^t must have same eigenvalues.

(8) The product of the eigenvalues of A is equal to the determinant of A.

(e) Let A be an $n \times n$ invertible matrix and λ is an eigenvalue of A. Which of the following is correct?

(1) $a\lambda$ is an eigenvalue of aA for any real number a.

(2) λ^n is an eigenvalue of A^n.

(4) λ^{-1} is an eigenvalue of A^{-1}.

(8) λ is an eigenvalue of A^t.

(f) Which of the following operations will not change the determinant of any square matrix?

(1) interchanging two columns.

(2) multiplying a column by a non-zero constant.

(4) adding a multiple of a row to another row.

(8) adding a multiple of a column to another column.

(g) If A and B are both $n \times n$ matrices with rank k, which of the following must be true?

(1) $\text{rank}(A + B) = k$.

(2) $\text{rank}(A + B) \leq k$.

(4) $\text{rank}(A + B) \geq k$.

(8) $\text{rank}(A + B) = \text{rank}(A - B)$.

2. (9 marks) Evaluate

$$\iint_R \exp\left(-\frac{x^2}{a^2} - \frac{y^2}{b^2}\right) dydx$$

where R is the quarter ellipse

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right) \leq 1, \ x \geq 0, \ y \geq 0.$$

3. (10 marks) Find the volume of the set enclosed by the surfaces $z = (1 + |x| + |y|)^{1/2}$, $z = -(1 + |x| + |y|)^{1/2}$ and by the planes $x = 1$, $x = -1$, $y = 1$ and $y = -1$.

4. (10 marks) Find all critical points of the function $f(x, y) = (ax^2 + by^2)e^{-x^2-y^2}$ $(a > b > 0)$. Determine the nature of these critical points (relative maximum, relative minimum or saddle point).

5. (10 marks) Determine whether the function

$$f(x, y) = \begin{cases} \dfrac{\sin(xy)}{x+y} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0). \end{cases}$$

is continuous at $(0, 0)$ or not. Prove your result.

6. (10 marks) Find the area of the overlap region of the cardioid $r = 1 + \sin\theta$ and the circle $r = 3\sin\theta$.(Keep exact solution.)

7. (10 marks)

   (a) (6 marks) If the set $\{\vec{u}, \vec{v}, \vec{w}\}$ is a linearly independent set, how about the set $\{\vec{u}, \vec{u} + \vec{v}, \vec{u} + \vec{v} + \vec{w}\}$? Is it linearly dependent or linearly independent? Explain.

   (b) (4 marks) If the set $\{\vec{u}, \vec{v}, \vec{w}\}$ is a linearly independent set, how about the set $\{\vec{u} - \vec{v}, \vec{v} - \vec{w}, \vec{w} - \vec{u}\}$? Is it linearly dependent or linearly independent? Explain.

8. (10 marks) Given that

$$A = \begin{bmatrix} 2 & 2-a & 2-a \\ 0 & 4-a & 2-a \\ 0 & -4+2a & -2+2a \end{bmatrix}$$

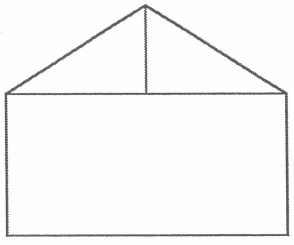
where $a \neq 2$.

(a) (5 marks) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) (2 marks) Find a non-singular matrix P and a diagonal matrix D, such that $A = PDP^{-1}$.

(c) (3 marks) Find the $(3,3)$-entry of A^{2022}.

9. (10 marks) Suppose a pentagon is composed of a rectangle surmounted by an isosceles triangle. If the length of the outer perimeter P is fixed, find the maximum possible area of the pentagon.



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